

Mass Tree Model

5.4

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Module Index

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Chapter 2

Namespace Index

2.1 Namespace List

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Data Structure Index

4.1 Data Structures

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Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

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mass.cc	Define methods for the MassBody class	81
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mass_body_links.hh	Define the class MassBodyLinks, the class that encapsulates the links between mass bodies .	83
mass_calc_composite_cm.cc	Calculate the aggregate cm for a composite MassBody	84
mass_calc_composite_inertia.cc	Calculate the aggregate inertia matrix for a composite MassBody	84
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mass_point_mass_inertia.cc	Compute the inertia tensor of a point mass	89
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Chapter 6

Module Documentation

6.1 Models

Modules

- [Dynamics](#)

6.1.1 Detailed Description

6.2 Dynamics

Modules

- [Mass](#)

6.2.1 Detailed Description

6.3 Mass

Files

- file [class_declarations.hh](#)
Forward declarations of mass model classes.
- file [mass.hh](#)
Define the class MassBody.
- file [mass_body_links.hh](#)
Define the class MassBodyLinks, the class that encapsulates the links between mass bodies.
- file [mass_messages.hh](#)
Define the class MassBodyMessages, the class that specifies the message IDs used in the MassBody model.
- file [mass_point.hh](#)
Define the class MassPoint, which defines the base features of a point related to a MassBody.
- file [mass_point_init.hh](#)

- Define the class `MassPointInit`, which initializes a `MassPoint` object.

 - file [mass_point_links.hh](#)

Define the class `MassPointLinks`, the class that encapsulates the links between mass points.
 - file [mass_point_state.hh](#)

Define the class `MassPointState`, which defines the state – position and orientation – of a `MassPoint`.
 - file [mass_properties.hh](#)

Define the class `MassProperties`.
 - file [mass_properties_init.hh](#)

Define class `MassPropertiesInit`, which initializes a mass properties object.
 - file [mass.cc](#)

Define methods for the `MassBody` class.
 - file [mass_attach.cc](#)

Define `MassBody` attach methods.
 - file [mass_calc_composite_cm.cc](#)

Calculate the aggregate cm for a composite `MassBody`.
 - file [mass_calc_composite_inertia.cc](#)

Calculate the aggregate inertia matrix for a composite `MassBody`.
 - file [mass_detach.cc](#)

Define `MassBody` detachment methods.
 - file [mass_messages.cc](#)

Implement the class `MassBodyMessages`.
 - file [mass_point.cc](#)

Define basic methods for the `MassPoint` class.
 - file [mass_point_init.cc](#)

Define methods for the `MassPointInit` class.
 - file [mass_point_mass_inertia.cc](#)

Compute the inertia tensor of a point mass.
 - file [mass_point_state.cc](#)

Define basic methods for the `MassPointState` class.
 - file [mass_print_body.cc](#)

Define `MassBody::print_body`.
 - file [mass_print_tree.cc](#)

Print out the `MassTree` members' data to a given file.
 - file [mass_properties_init.cc](#)

Define methods for the `MassPropertiesInit` class.
 - file [mass_reattach.cc](#)

Define `MassBody::reattach`.
 - file [mass_update.cc](#)

Define `MassBody::update_mass_properties`.

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [MassBody](#)
Represent both an atomic chunk of mass and an interconnected set of masses.
- class [MassBodyLinks](#)
Encapsulates the links between mass bodies.
- class [MassBodyMessages](#)
Specify the message IDs used in the [MassBody](#) model.
- class [MassPoint](#)
Adds tree linkages and a name to a [MassPointState](#).
- class [MassPointInit](#)
Contains data used to initialize a [MassPoint](#).
- class [MassPointLinks](#)
Encapsulates the links between mass points.
- class [MassPointState](#)
Defines the state – position and orientation – of a [MassPoint](#).
- class [MassProperties](#)
Defines mass properties – mass and inertia tensor.
- class [MassPropertiesInit](#)
Contains data used to initialize a mass model object.

Functions

- static void [generate_bad_point_message](#) (const char *file, unsigned int line, const std::string &child_body↔
_name, const std::string &child_point_name, const [MassPoint](#) *child_point, const std::string &parent_body↔
_name, const std::string &parent_point_name, const [MassPoint](#) *parent_point)
Generate a message regarding failure to find mass points on a vehicle.

7.1.1 Detailed Description

Namespace jeod.

7.1.2 Function Documentation

7.1.2.1 generate_bad_point_message()

```
static void jeod::generate_bad_point_message (
    const char * file,
    unsigned int line,
    const std::string & child_body_name,
    const std::string & child_point_name,
    const MassPoint * child_point,
    const std::string & parent_body_name,
    const std::string & parent_point_name,
    const MassPoint * parent_point ) [static]
```

Generate a message regarding failure to find mass points on a vehicle.

Assumptions and Limitations

- One or both of the input [MassPoint](#) pointers is null.

Parameters

in	<i>file</i>	File name
in	<i>line</i>	Line number
in	<i>child_body_name</i>	Name of child body
in	<i>child_point_name</i>	Name of child mass point
in	<i>child_point</i>	Child mass point
in	<i>parent_body_name</i>	Name of parent body
in	<i>parent_point_name</i>	Name of parent mass point
in	<i>parent_point</i>	Parent mass point

Definition at line 543 of file mass_attach.cc.

References `jeod::MassBodyMessages::invalid_attach`.

Referenced by `jeod::MassBody::attach_to()`.

Chapter 8

Data Structure Documentation

8.1 jeod::MassBody Class Reference

Represent both an atomic chunk of mass and an interconnected set of masses.

```
#include <mass.hh>
```

Public Member Functions

- [MassBody](#) ()
Default constructor; constructs a [MassBody](#) object.
- virtual [~MassBody](#) ()
Destroy a [MassBody](#) object.
- [MassBody](#) (const [MassBody](#) &)=delete
- void [operator=](#) (const [MassBody](#) &)=delete
- void [initialize_mass](#) (const [MassPropertiesInit](#) &properties, const std::vector< [MassPointInit](#) * > &points)
Initialize a [MassBody](#) object.
- bool [get_mass_properties_initialized](#) ()
- void [set_name](#) (std::string name_in)
- virtual bool [attach_to](#) (const std::string &this_point_name, const std::string &parent_point_name, [MassBody](#) &parent)
Attach this mass body's root body as a child of the specified mass body such that the specified mass points on the two bodies are coincident and the frames associated with those mass points are related by a 180 degree yaw.
- virtual bool [attach_to](#) (double offset_pstr_cstr_pstr[3], double T_pstr_cstr[3][3], [MassBody](#) &parent)
Attach this mass body's root body as a child of the specified mass body such that this body's structural origin is offset from the parent body's structural origin and this body's structural axes are oriented with respect to the parent body's structural axes as specified.
- virtual bool [attach_child](#) (const std::string &this_point_name, const std::string &child_point_name, [MassBody](#) &child)
Attach a child [MassBody](#) by point specification.
- virtual bool [attach_child](#) (double offset_pstr_cstr_pstr[3], double T_pstr_cstr[3][3], [MassBody](#) &child)
Attach a child [MassBody](#) by location specification.
- virtual bool [attach_validate](#) (const [MassBody](#) &parent, bool generate_message) const
Validate whether the pending attachment is legal.
- virtual bool [detach](#) ([MassBody](#) &from_body)

Detach the two bodies, 'this' and the argument body, from each other such that the detachment occurs between the superior body and the immediate child along the path from the inferior body to the superior body.

- virtual bool `detach` ()
Detach a mass body from its immediate parent.
- virtual bool `detach_validate` (const `MassBody` *parent, bool generate_message) const
Validate whether the pending detachment is legal.
- virtual bool `reattach` (double offset[3], double T_pstr_cstr[3][3])
Re-attach a child `MassBody` to a parent `MassBody`.
- const `MassPoint` * `find_mass_point` (const std::string &pt_name) const
Find the mass point with the given name.
- virtual void `add_mass_point` (const `MassPointInit` &mass_point_init)
Add a mass point to the list of such.
- std::size_t `mass_points_size` () const
Return the number of mass points for this body.
- bool `is_progeny_of` (const `MassBody` &test_body) const
Return true if this `MassBody` is an offspring of provided one, false if not.
- virtual const `MassBody` * `get_parent_body` () const
Returns the `MassBody`'s parent body, as a const pointer.
- virtual const `MassBody` * `get_root_body` () const
Finds & returns root of current `MassBody`'s tree.
- void `set_update_flag` ()
Flag mass bodies from the current body on up the mass tree as in need of mass property updates.
- virtual void `update_mass_properties` ()
Update composite mass properties for a mass tree marked for update.
- void `print_body` (FILE *file_ptr, int levels) const
Recursively print out the data in this and its children to given file.
- void `print_tree` (const char *file_name, int levels) const
Print out the `MassTree` members' data to a given file.

Static Public Member Functions

- static void `compute_point_mass_inertia` (double mass, const double r_pt[3], double inertia[3][3])
Compute the inertia tensor of a point mass.

Data Fields

- NamedItem `name`
Body name.
- `MassProperties` `core_properties`
The mass properties of this body alone, excluding child bodies.
- `MassProperties` `composite_properties`
The mass properties of this body, including child bodies.
- bool `compute_inverse_inertia` {}
When clear, the inverse of the inertia tensor is not calculated.
- `MassPoint` `structure_point`
The location and orientation of the structural frame with respect to the parent body's structural frame.
- `DynBody` *const `dyn_owner` {}
Indicates this body belongs to a `DynBody`.

Protected Member Functions

- [MassBody](#) ([DynBody](#) &owner)
Create a [MassBody](#), marking `is_dynamic` as true.
- virtual bool [attach_root_body](#) (double offset_pstr_cstr_pstr[3], double T_pstr_cstr[3][3], [MassBody](#) &parent)
Attach this mass body, which must be a root body, as a child of the specified mass body such that this body's structural origin is offset from the parent body's structural origin and this body's structural axes are oriented with respect to the parent body's structural axes as specified.
- virtual bool [attach_validate_parent](#) (const [MassBody](#) &parent, bool generate_message) const
Validate whether the pending attachment is legal from a connectivity point of view.
- virtual bool [attach_validate_child](#) (const [MassBody](#) &child, bool generate_message) const
Validate whether the pending attachment is legal from a physical point of view.
- virtual void [attach_establish_links](#) ([MassBody](#) &parent)
Establish the logical connectivity between parent and child.
- virtual void [attach_update_properties](#) (const double offset_pstr_cstr_pstr[3], const double T_pstr_cstr[3][3], [MassBody](#) &child)
Set the relation between parent and child and update the mass properties.
- virtual bool [detach_validate_parent](#) (const [MassBody](#) *parent, bool generate_message) const
Validate whether the pending detachment is legal from a connectivity point of view.
- virtual bool [detach_validate_child](#) (const [MassBody](#) &child, bool generate_message) const
Validate whether the pending detachment is legal from a mass tree point of view.
- virtual void [detach_sever_links](#) ([MassBody](#) &parent)
Break the logical connectivity between parent and child.
- virtual void [detach_update_properties](#) ([MassBody](#) &child)
Update parent and child properties to reflect that they are detached.
- virtual [MassBody](#) * [get_parent_body_internal](#) ()
Returns the [MassBody](#)'s parent body, as a non-const pointer.
- virtual [MassBody](#) * [get_root_body_internal](#) ()
Finds & returns root of current [MassBody](#)'s tree.
- void [calc_composite_cm](#) ()
Calculate the aggregate cm for a composite [MassBody](#).
- void [calc_composite_inertia](#) ()
Calculate the aggregate inertia matrix for a composite [MassBody](#).

Protected Attributes

- BaseDynManager * [dyn_manager](#) {}
The dynamics manager for the simulation.
- bool [mass_properties_initialized](#) {}
Initialized flag.
- [MassBodyLinks](#) links
Linkage to rest of mass tree.
- bool [needs_update](#) {}
When set, indicates that an update to the composite mass is needed.
- std::list< [MassPoint](#) * > [mass_points](#)
List of points associated with this mass body.
- [MassPoint](#) [core_wrt_composite](#)
The location and orientation of the composite CoM and body frame with respect to the parent body's composite CoM and body frame.
- [MassPoint](#) [composite_wrt_pstr](#)
The location and orientation of the composite CoM and body frame with respect to the parent body's structural frame.
- [MassPoint](#) [composite_wrt_pbdy](#)
The location and orientation of the composite CoM and body frame with respect to the parent body's composite CoM and body frame.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2f](#) (jeod, [MassBody](#))

Friends

- class [MassBodyLinks](#)
- class [DynBody](#)

8.1.1 Detailed Description

Represent both an atomic chunk of mass and an interconnected set of masses.

Multiple [MassBody](#) objects can be attached to one another in a tree structure. This connected sets of bodies has a composite set of mass properties. Sans these connections, a body will have some core mass proprties.

The [MassBody](#) class is one of the key classes in JEOD 2.0. In addition to representing masses, it is also the parent of the [DynBody](#) class, which is used in JEOD 2.0 to represent vehicles.

Assumptions and Limitations

- Rigid Bodies
- Tree attachment Structure

Definition at line 112 of file mass.hh.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 [MassBody\(\)](#) [1/3]

```
jeod::MassBody::MassBody ( )
```

Default constructor; constructs a [MassBody](#) object.

Definition at line 76 of file mass.cc.

8.1.2.2 [~MassBody\(\)](#)

```
jeod::MassBody::~~MassBody ( ) [virtual]
```

Destroy a [MassBody](#) object.

Definition at line 140 of file mass.cc.

References [detach\(\)](#), [jeod::MassPoint::detach\(\)](#), [jeod::MassBodyMessages::invalid_detach](#), [links](#), [mass_points](#), [name](#), and [set_update_flag\(\)](#).

8.1.2.3 MassBody() [2/3]

```
jeod::MassBody::MassBody (
    const MassBody & ) [delete]
```

8.1.2.4 MassBody() [3/3]

```
jeod::MassBody::MassBody (
    DynBody & owner ) [explicit], [protected]
```

Create a [MassBody](#), marking `is_dynamic` as true.

Default constructor; constructs a [MassBody](#) object.

Utilized by [DynBody](#) via friendship.

Definition at line 107 of file `mass.cc`.

References [jeod::MassPoint::attach\(\)](#), [composite_properties](#), [composite_wrt_pbdy](#), [core_properties](#), [core_wrt_](#)↵
[composite](#), [structure_point](#), and [jeod::MassPointState::T_parent_this](#).

8.1.3 Member Function Documentation

8.1.3.1 add_mass_point()

```
void jeod::MassBody::add_mass_point (
    const MassPointInit & mass_point_init ) [virtual]
```

Add a mass point to the list of such.

Parameters

<code>in</code>	<code>mass_point_init</code>	Mass point spec
-----------------	------------------------------	-----------------

Definition at line 339 of file `mass.cc`.

References [jeod::MassPoint::attach\(\)](#), [find_mass_point\(\)](#), [jeod::MassPointInit::initialize_mass_point\(\)](#), [jeod::](#)↵
[MassBodyMessages::invalid_name](#), [mass_points](#), [name](#), [jeod::MassPoint::name](#), [jeod::MassPointInit::name](#), and
[structure_point](#).

Referenced by [initialize_mass\(\)](#).

8.1.3.2 attach_child() [1/2]

```
bool jeod::MassBody::attach_child (
    const std::string & this_point_name,
    const std::string & child_point_name,
    MassBody & child ) [virtual]
```

Attach a child [MassBody](#) by point specification.

See corresponding [MassBody::attach_to\(\)](#) method for more information.

Definition at line 231 of file `mass_attach.cc`.

References [attach_to\(\)](#).

8.1.3.3 attach_child() [2/2]

```
bool jeod::MassBody::attach_child (
    double offset_pstr_cstr_pstr[3],
    double T_pstr_cstr[3][3],
    MassBody & child ) [virtual]
```

Attach a child [MassBody](#) by location specification.

See corresponding [MassBody::attach_to\(\)](#) method for more information. Note that the offset and transformation are specified w.r.t. the parent in both [attach_to\(\)](#) and [attach_child\(\)](#)

Definition at line 244 of file `mass_attach.cc`.

References [attach_to\(\)](#).

8.1.3.4 attach_establish_links()

```
void jeod::MassBody::attach_establish_links (
    MassBody & parent ) [protected], [virtual]
```

Establish the logical connectivity between parent and child.

Extensibility comments –

- This method is invoked before the computing the physical relation between parent and child.
- The generic purpose of this method is to establish the logical connectivity between parent and child in terms of the child class.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The attachment is valid; not checked.

Parameters

<i>in, out</i>	<i>parent</i>	The new parent body; the body to which this body is to be attached.
----------------	---------------	---

Definition at line 461 of file mass_attach.cc.

References `jeod::MassPoint::attach()`, `composite_wrt_pbdy`, `composite_wrt_pstr`, `links`, and `structure_point`.

Referenced by `attach_root_body()`.

8.1.3.5 attach_root_body()

```
bool jeod::MassBody::attach_root_body (
    double offset_pstr_cstr_pstr[3],
    double T_pstr_cstr[3][3],
    MassBody & parent ) [protected], [virtual]
```

Attach this mass body, which must be a root body, as a child of the specified mass body such that this body's structural origin is offset from the parent body's structural origin and this body's structural axes are oriented with respect to the parent body's structural axes as specified.

Assumptions and Limitations

- The subject body, this, must be a root body.

Returns

Success indicator: true=success, false=attachment not performed.

Parameters

<i>in</i>	<i>offset_pstr_cstr_pstr</i>	Location of this body's structural origin with respect to the new parent body's structural origin, specified in structural coordinates of the new parent body. Units: M
<i>in</i>	<i>T_pstr_cstr</i>	Transformation matrix from the new parent body's structural frame to this body's structural frame.
<i>in, out</i>	<i>parent</i>	The new parent body; the body to which this body is to be attached.

Definition at line 267 of file mass_attach.cc.

References `attach_establish_links()`, `attach_update_properties()`, `attach_validate()`, `jeod::MassBodyMessages::invalid_attach`, `links`, and `name`.

Referenced by `attach_to()`.

8.1.3.6 attach_to() [1/2]

```
bool jeod::MassBody::attach_to (
    const std::string & this_point_name,
    const std::string & parent_point_name,
    MassBody & parent ) [virtual]
```

Attach this mass body's root body as a child of the specified mass body such that the specified mass points on the two bodies are coincident and the frames associated with those mass points are related by a 180 degree yaw.

Returns

Success indicator: true=success, false=attachment not performed.

Parameters

in	<i>this_point_name</i>	The name of a mass point contained in this mass body's list of mass points.
in	<i>parent_point_name</i>	The name of a mass point contained in the new parent body's list of mass points.
in, out	<i>parent</i>	The new parent body; the body to which this body's root body is to be attached.

Definition at line 67 of file mass_attach.cc.

References jeod::MassPoint::attach(), attach_validate(), jeod::MassPoint::compute_state_wrt_pred(), jeod::MassPoint::detach(), find_mass_point(), jeod::generate_bad_point_message(), name, jeod::MassPointState::position, jeod::MassPointState::Q_parent_this, structure_point, and jeod::MassPointState::T_parent_this.

Referenced by attach_child().

8.1.3.7 attach_to() [2/2]

```
bool jeod::MassBody::attach_to (
    double offset_pstr_cstr_pstr[3],
    double T_pstr_cstr[3][3],
    MassBody & parent ) [virtual]
```

Attach this mass body's root body as a child of the specified mass body such that this body's structural origin is offset from the parent body's structural origin and this body's structural axes are oriented with respect to the parent body's structural axes as specified.

Returns

Success indicator: true=success, false=attachment not performed.

Parameters

in	<i>offset_pstr_cstr_pstr</i>	Location of this body's structural origin with respect to the new parent body's structural origin, specified in structural coordinates of the new parent body. Units: M
in	<i>T_pstr_cstr</i>	Transformation matrix from the new parent body's structural frame to this body's structural frame.
in, out	<i>parent</i>	The new parent body; the body to which this body's root body is to be attached.

Definition at line 151 of file mass_attach.cc.

References `jeod::MassBodyMessages::attach_info`, `attach_root_body()`, `attach_validate()`, `jeod::MassPointState::compute_state_wrt_pred()`, `get_root_body()`, `get_root_body_internal()`, `jeod::MassBodyMessages::invalid_attach`, `links`, `name`, `jeod::MassPointState::position`, `structure_point`, and `jeod::MassPointState::T_parent_this`.

8.1.3.8 attach_update_properties()

```
void jeod::MassBody::attach_update_properties (
    const double offset_pstr_cstr_pstr[3],
    const double T_pstr_cstr[3][3],
    MassBody & child ) [protected], [virtual]
```

Set the relation between parent and child and update the mass properties.

Extensibility comments –

- This method is sent to the parent body of the attachment after the child body has established the logical connectivity between the parent body and child body.
- The generic purpose of this method is to establish the physical relation between parent and child and to update any physical properties that change as a result of the attachment.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The attachment is valid and logical connectivity has been established. Neither assumption is checked.

Parameters

in	<i>offset_pstr_cstr_pstr</i>	Location of this body's structural origin with respect to the new parent body's structural origin, specified in structural coordinates of the new parent body. Units: M
in	<i>T_pstr_cstr</i>	Transformation matrix from the new parent body's structural frame to this body's structural frame.
in, out	<i>child</i>	The child body; the body newly attached to this body.

Definition at line 494 of file mass_attach.cc.

References composite_properties, composite_wrt_pbdy, composite_wrt_pstr, jeod::MassPointState::compute_transformation(), get_root_body_internal(), jeod::MassPointState::position, jeod::MassPointState::Q_parent_this, set_update_flag(), structure_point, jeod::MassPointState::T_parent_this, update_mass_properties(), jeod::MassPointState::update_orientation(), and jeod::MassPointState::update_point().

Referenced by attach_root_body().

8.1.3.9 attach_validate()

```
bool jeod::MassBody::attach_validate (
    const MassBody & parent,
    bool generate_message ) const [virtual]
```

Validate whether the pending attachment is legal.

Extensibility comments –

- Overriding this method doesn't make a whole lot of sense.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The subject body, this, must be a root body. This is not checked.

Returns

Validity indicator

Parameters

in	<i>parent</i>	The new parent body; the body to which this body is to be attached.
in	<i>generate_message</i>	Generate message if invalid?

Definition at line 321 of file mass_attach.cc.

References attach_validate_child(), and attach_validate_parent().

Referenced by attach_root_body(), and attach_to().

8.1.3.10 attach_validate_child()

```
bool jeod::MassBody::attach_validate_child (
    const MassBody & child,
    bool generate_message ) const [protected], [virtual]
```

Validate whether the pending attachment is legal from a physical point of view.

Extensibility comments –

- This method determines whether invoking `attach_update_properties` makes sense.
- This is a free pass for a [MassBody](#).

Returns

Validity indicator

Parameters

in	<i>child</i>	The child body; the body to be attached to this body.
in	<i>generate_message</i>	Generate message if invalid?

Definition at line 405 of file `mass_attach.cc`.

References `dyn_manager`, `get_root_body()`, `jeod::MassBodyMessages::invalid_attach`, and `name`.

Referenced by `attach_validate()`.

8.1.3.11 attach_validate_parent()

```
bool jeod::MassBody::attach_validate_parent (
    const MassBody & parent,
    bool generate_message ) const [protected], [virtual]
```

Validate whether the pending attachment is legal from a connectivity point of view.

Extensibility comments –

- This method determines whether invoking `attach_establish_links` makes sense.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The subject body, this, must be a root body. This is not checked.

Returns

Validity indicator

Parameters

in	<i>parent</i>	The new parent body; the body to which this body is to be attached.
in	<i>generate_message</i>	Generate message if invalid?

Definition at line 350 of file mass_attach.cc.

References dyn_manager, get_root_body(), jeod::MassBodyMessages::invalid_attach, and name.

Referenced by attach_validate().

8.1.3.12 calc_composite_cm()

```
void jeod::MassBody::calc_composite_cm ( ) [protected]
```

Calculate the aggregate cm for a composite [MassBody](#).

Assumptions and Limitations

- Rigid bodies
- Must calculate from bottom to top of tree for meaningful results

Definition at line 45 of file mass_calc_composite_cm.cc.

References composite_properties, composite_wrt_pstr, core_properties, jeod::MassProperties::inverse_mass, links, jeod::MassProperties::mass, and jeod::MassPointState::position.

Referenced by update_mass_properties().

8.1.3.13 calc_composite_inertia()

```
void jeod::MassBody::calc_composite_inertia ( ) [protected]
```

Calculate the aggregate inertia matrix for a composite [MassBody](#).

Assumptions and Limitations

- Rigid Bodies
- Tree attachment structure
- Must calculate from bottom to top of tree for meaningful results

Definition at line 45 of file mass_calc_composite_inertia.cc.

References composite_properties, composite_wrt_pbdy, compute_point_mass_inertia(), core_properties, core←_wrt_composite, jeod::MassProperties::inertia, links, jeod::MassProperties::mass, jeod::MassPointState::position, and jeod::MassPointState::T_parent_this.

Referenced by update_mass_properties().

8.1.3.14 compute_point_mass_inertia()

```
void jeod::MassBody::compute_point_mass_inertia (
    double mass,
    const double r_pt[3],
    double inertia[3][3] ) [static]
```

Compute the inertia tensor of a point mass.

Parameters

in	<i>mass</i>	Mass of point mass Units: kg
in	<i>r_pt</i>	Vector to point mass Units: M
out	<i>inertia</i>	Inertia tensor Units: kgM2

Definition at line 41 of file mass_point_mass_inertia.cc.

Referenced by calc_composite_inertia(), and jeod::MassPropertiesInit::initialize_mass_properties().

8.1.3.15 detach() [1/2]

```
bool jeod::MassBody::detach ( ) [virtual]
```

Detach a mass body from its immediate parent.

Returns true to indicate success, false to indicate failure.

Assumptions and Limitations

- Instances of more derived classes, with presumably more involved dynamics, are situated higher in the mass tree than are more basic instances. For example, a simple [MassBody](#) can be a child of a [DynBody](#), but not the other way around.
- The attachment in the mass tree between the immediate child and the superior body is assumed to reflect a real physical attachment.

Returns

Success flag

Definition at line 133 of file mass_detach.cc.

References detach_sever_links(), detach_update_properties(), detach_validate(), dyn_owner, DynBody, and links.

Referenced by ~MassBody().

8.1.3.16 detach() [2/2]

```
bool jeod::MassBody::detach (
    MassBody & mass_body ) [virtual]
```

Detach the two bodies, 'this' and the argument body, from each other such such that the detachment occurs between the superior body and the immediate child along the path from the inferior body to the superior body.

Returns true to indicate success, false to indicate failure.

Assumptions and Limitations

- Instances of more derived classes, with presumably more involved dynamics, are situated higher in the mass tree than are more basic instances. For example, a simple [MassBody](#) can be a child of a [DynBody](#), but not the other way around.
- The attachment in the mass tree between the immediate child and the superior body is assumed to reflect a real physical attachment.

Returns

Success flag

Parameters

<i>in, out</i>	<i>mass_body</i>	The other body
----------------	------------------	----------------

Definition at line 63 of file mass_detach.cc.

References detach(), jeod::MassBodyMessages::invalid_detach, links, and name.

Referenced by detach().

8.1.3.17 detach_sever_links()

```
void jeod::MassBody::detach_sever_links (
    MassBody & parent ) [protected], [virtual]
```

Break the logical connectivity between parent and child.

Extensibility comments –

- This method is invoked before the updating the parent/child states.
- The generic purpose of this method is to sever all connectivity links between parent and child, in terms of the child class.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The detachment is valid; not checked.

Parameters

<i>in, out</i>	<i>parent</i>	The parent body; the body from which this body is to be detached.
----------------	---------------	---

Definition at line 292 of file mass_detach.cc.

References links.

Referenced by detach().

8.1.3.18 detach_update_properties()

```
void jeod::MassBody::detach_update_properties (
    MassBody & child ) [protected], [virtual]
```

Update parent and child properties to reflect that they are detached.

Extensibility comments –

- This method is sent to the parent body of the detachment after the child body has severed the logical connectivity between the parent body and child body.
- The generic purpose of this method is to update any physical properties that change as a result of the detachment.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The detachment is valid and logical connectivity has been severed. Neither assumption is checked.

Parameters

<i>in, out</i>	<i>child</i>	The child body; the body newly detached from this body.
----------------	--------------	---

Definition at line 315 of file mass_detach.cc.

References `composite_properties`, `composite_wrt_pbdy`, `composite_wrt_pstr`, `compute_inverse_inertia`, `get_root_body_internal()`, `jeod::MassProperties::inertia`, `jeod::MassPoint::initialize_mass_point()`, `jeod::MassProperties::inverse_inertia`, `jeod::MassProperties::mass`, `set_update_flag()`, `structure_point`, and `update_mass_properties()`.

Referenced by `detach()`.

8.1.3.19 detach_validate()

```
bool jeod::MassBody::detach_validate (
    const MassBody * parent,
    bool generate_message ) const [virtual]
```

Validate whether the pending detachment is legal.

Extensibility comments –

- Overriding this method doesn't make a whole lot of sense.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Assumptions and Limitations

- The subject body, this, must be a root body. This is not checked.

Returns

Validity indicator

Parameters

<i>in</i>	<i>parent</i>	The parent body; the body from which this body is to be detached.
<i>in</i>	<i>generate_message</i>	Generate message if invalid?

Generated by Doxygen

Definition at line 184 of file mass_detach.cc.

References detach_validate_child(), and detach_validate_parent().

Referenced by detach().

8.1.3.20 detach_validate_child()

```
bool jeod::MassBody::detach_validate_child (
    const MassBody & child,
    bool generate_message ) const [protected], [virtual]
```

Validate whether the pending detachment is legal from a mass tree point of view.

Extensibility comments –

- This method determines whether invoking detach_update_properties makes sense.

Returns

Validity indicator

Parameters

in	<i>child</i>	The child body; the body to be detached from this body.
in	<i>generate_message</i>	Generate message if invalid?

Definition at line 257 of file mass_detach.cc.

References dyn_manager, jeod::MassBodyMessages::invalid_detach, and name.

Referenced by detach_validate().

8.1.3.21 detach_validate_parent()

```
bool jeod::MassBody::detach_validate_parent (
    const MassBody * parent,
    bool generate_message ) const [protected], [virtual]
```

Validate whether the pending detachment is legal from a connectivity point of view.

Extensibility comments –

- This method determines whether invoking detach_sever_links makes sense.
- Any class that overrides this method must either invoke this method or perform the actions performed herein.

Returns

Validity indicator

Parameters

in	<i>parent</i>	The parent body; the body from which this body is to be detached.
in	<i>generate_message</i>	Generate message if invalid?

Definition at line 210 of file mass_detach.cc.

References dyn_manager, jeod::MassBodyMessages::invalid_detach, and name.

Referenced by detach_validate().

8.1.3.22 find_mass_point()

```
const MassPoint * jeod::MassBody::find_mass_point (
    const std::string & pt_name ) const
```

Find the mass point with the given name.

Returns

Mass point

Parameters

in	<i>pt_name</i>	mass point name
----	----------------	-----------------

Definition at line 307 of file mass.cc.

References mass_points, and name.

Referenced by add_mass_point(), and attach_to().

8.1.3.23 get_mass_properties_initialized()

```
bool jeod::MassBody::get_mass_properties_initialized ( ) [inline]
```

Definition at line 134 of file mass.hh.

References mass_properties_initialized.

8.1.3.24 `get_parent_body()`

```
const MassBody * jeod::MassBody::get_parent_body ( ) const [virtual]
```

Returns the [MassBody](#)'s parent body, as a const pointer.

Returns

Pointer to parent body

Definition at line 224 of file mass.cc.

References links.

8.1.3.25 `get_parent_body_internal()`

```
MassBody * jeod::MassBody::get_parent_body_internal ( ) [protected], [virtual]
```

Returns the [MassBody](#)'s parent body, as a non-const pointer.

Returns

Pointer to parent body

Definition at line 233 of file mass.cc.

References links.

8.1.3.26 `get_root_body()`

```
const MassBody * jeod::MassBody::get_root_body ( ) const [virtual]
```

Finds & returns root of current [MassBody](#)'s tree.

Returns

Pointer to root body

Definition at line 242 of file mass.cc.

References links.

Referenced by `attach_to()`, `attach_validate_child()`, `attach_validate_parent()`, and `print_tree()`.

8.1.3.27 get_root_body_internal()

```
MassBody * jeod::MassBody::get_root_body_internal ( ) [protected], [virtual]
```

Finds & returns root of current [MassBody](#)'s tree.

Returns

Pointer to root body

Definition at line 251 of file mass.cc.

References links.

Referenced by [attach_to\(\)](#), [attach_update_properties\(\)](#), [detach_update_properties\(\)](#), and [reattach\(\)](#).

8.1.3.28 initialize_mass()

```
void jeod::MassBody::initialize_mass (
    const MassPropertiesInit & properties,
    const std::vector< MassPointInit * > & points )
```

Initialize a [MassBody](#) object.

Parameters

in	<i>properties</i>	Core mass ppty specs
in	<i>points</i>	Mass point specs

Definition at line 191 of file mass.cc.

References [add_mass_point\(\)](#), [composite_properties](#), [core_properties](#), [dyn_owner](#), [jeod::MassPropertiesInit::initialize_mass_properties\(\)](#), [mass_properties_initialized](#), [jeod::MassPointState::Q_parent_this](#), [set_update_flag\(\)](#), and [jeod::MassPointState::T_parent_this](#).

8.1.3.29 is_progeny_of()

```
bool jeod::MassBody::is_progeny_of (
    const MassBody & test_body ) const
```

Return true if this [MassBody](#) is an offspring of provided one, false if not.

Returns

Is this offspring of test_body?

Parameters

in	<i>test_body</i>	Other MassBody
----	------------------	--------------------------------

Definition at line 262 of file mass.cc.

References links.

8.1.3.30 JEOD_CLASS_ESTABLISH_FRIENDS2f()

```
jeod::MassBody::JEOD_CLASS_ESTABLISH_FRIENDS2f (
    jeod ,
    MassBody ) [private]
```

8.1.3.31 mass_points_size()

```
size_t jeod::MassBody::mass_points_size ( ) const
```

Return the number of mass points for this body.

Returns

Mass point

Definition at line 297 of file mass.cc.

References [mass_points](#).

8.1.3.32 operator=()

```
void jeod::MassBody::operator= (
    const MassBody & ) [delete]
```

8.1.3.33 print_body()

```
void jeod::MassBody::print_body (
    FILE * file_ptr,
    int levels ) const
```

Recursively print out the data in this and its children to given file.

Parameters

<i>in</i> , <i>out</i>	<i>file_ptr</i>	Output file stream
<i>in</i> , <i>out</i>	<i>levels</i>	Max desired recursion level

Definition at line 44 of file `mass_print_body.cc`.

References `composite_properties`, `core_properties`, `jeod::MassProperties::inertia`, `jeod::MassProperties::inverse_inertia`, `jeod::MassProperties::inverse_mass`, `links`, `jeod::MassProperties::mass`, `name`, `jeod::MassPointState::position`, `print_body()`, `structure_point`, and `jeod::MassPointState::T_parent_this`.

Referenced by `print_body()`, and `print_tree()`.

8.1.3.34 print_tree()

```
void jeod::MassBody::print_tree (
    const char * file_name,
    int levels ) const
```

Print out the MassTree members' data to a given file.

Parameters

<i>in</i>	<i>file_name</i>	Desired output file name
<i>in</i> , <i>out</i>	<i>levels</i>	Max desired recursion level

Definition at line 47 of file `mass_print_tree.cc`.

References `get_root_body()`, `jeod::MassBodyMessages::io_error`, and `print_body()`.

8.1.3.35 reattach()

```
bool jeod::MassBody::reattach (
    double offset[3],
    double T_pstr_cstr[3][3] ) [virtual]
```

Re-attach a child [MassBody](#) to a parent [MassBody](#).

Assumptions and Limitations

- Rigid Bodies
- Tree attachment structure
- Child is known to be a root or atomic body
- Re-establishing an attachment that previously existed

Parameters

in	<i>offset</i>	Desired offset from parent struct frame to attached child's struct frame, expressed in parent struct frame Units: M
in	<i>T_pstr_cstr</i>	Desired transformation matrix from parent struct frame to attached child's struct frame

Definition at line 52 of file mass_reattach.cc.

References `composite_properties`, `composite_wrt_pbdy`, `composite_wrt_pstr`, `jeod::MassPointState::compute_transformation()`, `get_root_body_internal()`, `jeod::MassBodyMessages::invalid_node`, `links`, `name`, `jeod::MassPointState::position`, `jeod::MassPointState::Q_parent_this`, `set_update_flag()`, `structure_point`, `jeod::MassPointState::T_parent_this`, `update_mass_properties()`, `jeod::MassPointState::update_orientation()`, and `jeod::MassPointState::update_point()`.

8.1.3.36 set_name()

```
void jeod::MassBody::set_name (
    std::string name_in ) [inline]
```

Definition at line 140 of file mass.hh.

References `name`.

8.1.3.37 set_update_flag()

```
void jeod::MassBody::set_update_flag ( )
```

Flag mass bodies from the current body on up the mass tree as in need of mass property updates.

Definition at line 284 of file mass.cc.

References `links`.

Referenced by `attach_update_properties()`, `detach_update_properties()`, `initialize_mass()`, `reattach()`, and `~MassBody()`.

8.1.3.38 update_mass_properties()

```
void jeod::MassBody::update_mass_properties ( ) [virtual]
```

Update composite mass properties for a mass tree marked for update.

The properties are updated from the bottom up to generate correct results.

Assumptions and Limitations

- Rigid bodies
- Tree attachment structure

Definition at line 50 of file mass_update.cc.

References `calc_composite_cm()`, `calc_composite_inertia()`, `composite_properties`, `composite_wrt_pbdy`, `composite_wrt_pstr`, `compute_inverse_inertia`, `core_properties`, `core_wrt_composite`, `jeod::MassProperties::inverse_inertia`, `jeod::MassProperties::inverse_inertia`, `jeod::MassProperties::inverse_mass`, `links`, `jeod::MassProperties::mass`, `needs_update`, `jeod::MassPointState::position`, `structure_point`, `jeod::MassPointState::T_parent_this`, and `update_mass_properties()`.

Referenced by `attach_update_properties()`, `detach_update_properties()`, `reattach()`, and `update_mass_properties()`.

8.1.4 Friends And Related Function Documentation

8.1.4.1 DynBody

```
friend class DynBody [friend]
```

Definition at line 115 of file mass.hh.

Referenced by `detach()`.

8.1.4.2 MassBodyLinks

```
friend class MassBodyLinks [friend]
```

Definition at line 114 of file mass.hh.

8.1.5 Field Documentation

8.1.5.1 composite_properties

`MassProperties jeod::MassBody::composite_properties`

The mass properties of this body, including child bodies.

The composite_properties are core_properties are synonymous for atomic bodies. The composite_properties is a child of the structure_point. trick_units(-)

Definition at line 227 of file mass.hh.

Referenced by attach_update_properties(), calc_composite_cm(), calc_composite_inertia(), detach_update_properties(), initialize_mass(), MassBody(), print_body(), reattach(), and update_mass_properties().

8.1.5.2 composite_wrt_pbdy

`MassPoint jeod::MassBody::composite_wrt_pbdy [protected]`

The location and orientation of the composite CoM and body frame with respect to the parent body's composite CoM and body frame.

trick_units(-)

Definition at line 356 of file mass.hh.

Referenced by attach_establish_links(), attach_update_properties(), calc_composite_inertia(), detach_update_properties(), MassBody(), reattach(), and update_mass_properties().

8.1.5.3 composite_wrt_pstr

`MassPoint jeod::MassBody::composite_wrt_pstr [protected]`

The location and orientation of the composite CoM and body frame with respect to the parent body's structural frame.

trick_units(-)

Definition at line 350 of file mass.hh.

Referenced by attach_establish_links(), attach_update_properties(), calc_composite_cm(), detach_update_properties(), reattach(), and update_mass_properties().

8.1.5.4 compute_inverse_inertia

```
bool jeod::MassBody::compute_inverse_inertia {}
```

When clear, the inverse of the inertia tensor is not calculated.

The inverse of the inertia tensor is only calculated if this flag is set and the composite mass is positive. `trick_units(-)`

Definition at line 234 of file `mass.hh`.

Referenced by `detach_update_properties()`, and `update_mass_properties()`.

8.1.5.5 core_properties

```
MassProperties jeod::MassBody::core_properties
```

The mass properties of this body alone, excluding child bodies.

The `core_properties` object is a child of the `structure_point`. `trick_units(-)`

Definition at line 220 of file `mass.hh`.

Referenced by `calc_composite_cm()`, `calc_composite_inertia()`, `initialize_mass()`, `MassBody()`, `print_body()`, and `update_mass_properties()`.

8.1.5.6 core_wrt_composite

```
MassPoint jeod::MassBody::core_wrt_composite [protected]
```

The location and orientation of the composite CoM and body frame with respect to the parent body's composite CoM and body frame.

`trick_units(-)`

Definition at line 344 of file `mass.hh`.

Referenced by `calc_composite_inertia()`, `MassBody()`, and `update_mass_properties()`.

8.1.5.7 dyn_manager

```
BaseDynManager* jeod::MassBody::dyn_manager {} [protected]
```

The dynamics manager for the simulation.

`trick_units(-)`

Definition at line 308 of file `mass.hh`.

Referenced by `attach_validate_child()`, `attach_validate_parent()`, `detach_validate_child()`, and `detach_validate_parent()`.

8.1.5.8 dyn_owner

```
DynBody* const jeod::MassBody::dyn_owner {}
```

Indicates this body belongs to a DynBody.

Many JEOD functions manipulate [MassBody](#) properties with the implementation depending/varying upon whether the [MassBody](#) refers to a dynamic body. `trick_units(-)`

Definition at line 249 of file `mass.hh`.

Referenced by `detach()`, and `initialize_mass()`.

8.1.5.9 links

```
MassBodyLinks jeod::MassBody::links [protected]
```

Linkage to rest of mass tree.

Programmatic interfaces:

- [MassBodyLinks](#) provides accessors to the parent and root and provides methods to attach, detach links (and hence bodies).
- This class provides accessors to the same.
- Various iterators provide the ability to iterate over child bodies and up the parent chain. `trick_units(-)`

Definition at line 326 of file `mass.hh`.

Referenced by `attach_establish_links()`, `attach_root_body()`, `attach_to()`, `calc_composite_cm()`, `calc_composite_inertia()`, `detach()`, `detach_sever_links()`, `get_parent_body()`, `get_parent_body_internal()`, `get_root_body()`, `get_root_body_internal()`, `is_progeny_of()`, `print_body()`, `reattach()`, `set_update_flag()`, `update_mass_properties()`, and `~MassBody()`.

8.1.5.10 mass_points

```
std::list<MassPoint *> jeod::MassBody::mass_points [protected]
```

List of points associated with this mass body.

NOTE WELL: The [MassBody](#) manages the memory associated with the contents of this list.

Definition at line 338 of file `mass.hh`.

Referenced by `add_mass_point()`, `find_mass_point()`, `mass_points_size()`, and `~MassBody()`.

8.1.5.11 mass_properties_initialized

```
bool jeod::MassBody::mass_properties_initialized {} [protected]
```

Initialized flag.

Indicates whether the initialize_mass method has been executed, specifically the initialize_mass_properties method from the initialize_mass method. trick_units(-)

Definition at line 315 of file mass.hh.

Referenced by get_mass_properties_initialized(), and initialize_mass().

8.1.5.12 name

```
NamedItem jeod::MassBody::name
```

Body name.

trick_units(-)

Definition at line 214 of file mass.hh.

Referenced by add_mass_point(), attach_root_body(), attach_to(), attach_validate_child(), attach_validate_parent(), detach(), detach_validate_child(), detach_validate_parent(), find_mass_point(), print_body(), reattach(), set_name(), and ~MassBody().

8.1.5.13 needs_update

```
bool jeod::MassBody::needs_update {} [protected]
```

When set, indicates that an update to the composite mass is needed.

trick_units(-)

Definition at line 331 of file mass.hh.

Referenced by update_mass_properties().

8.1.5.14 structure_point

`MassPoint jeod::MassBody::structure_point`

The location and orientation of the structural frame with respect to the parent body's structural frame.

Attaching one `MassBody` to some other `MassBody` makes the attaching `MassBody`'s `structure_point` a child of the attachee's `structure_point`. `trick_units(-)`

Definition at line 242 of file `mass.hh`.

Referenced by `add_mass_point()`, `attach_establish_links()`, `attach_to()`, `attach_update_properties()`, `detach_↔update_properties()`, `MassBody()`, `print_body()`, `reattach()`, and `update_mass_properties()`.

The documentation for this class was generated from the following files:

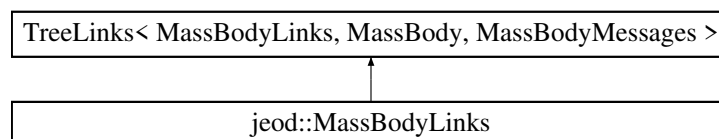
- `mass.hh`
- `mass.cc`
- `mass_attach.cc`
- `mass_calc_composite_cm.cc`
- `mass_calc_composite_inertia.cc`
- `mass_detach.cc`
- `mass_point_mass_inertia.cc`
- `mass_print_body.cc`
- `mass_print_tree.cc`
- `mass_reattach.cc`
- `mass_update.cc`

8.2 jeod::MassBodyLinks Class Reference

Encapsulates the links between mass bodies.

```
#include <mass_body_links.hh>
```

Inheritance diagram for `jeod::MassBodyLinks`:



Public Member Functions

- `MassBodyLinks (MassBody &container_in)`
Non-default constructor.
- `MassBodyLinks ()=delete`
- `MassBodyLinks (const MassBodyLinks &)=delete`
- `void operator= (const MassBodyLinks &)=delete`
- `~MassBodyLinks () override=default`
Destructor.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [MassBodyLinks](#))

Static Private Attributes

- static constexpr unsigned int [default_path_size](#) {8}

8.2.1 Detailed Description

Encapsulates the links between mass bodies.

Definition at line 81 of file `mass_body_links.hh`.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 MassBodyLinks() [1/3]

```
jeod::MassBodyLinks::MassBodyLinks (
    MassBody & container_in ) [inline], [explicit]
```

Non-default constructor.

Parameters

<i>container</i> ↔ _in	The MassBody object that contains this object.
---------------------------	--

Definition at line 89 of file `mass_body_links.hh`.

8.2.2.2 MassBodyLinks() [2/3]

```
jeod::MassBodyLinks::MassBodyLinks ( ) [delete]
```

8.2.2.3 MassBodyLinks() [3/3]

```
jeod::MassBodyLinks::MassBodyLinks (
    const MassBodyLinks & ) [delete]
```

8.2.2.4 ~MassBodyLinks()

```
jeod::MassBodyLinks::~~MassBodyLinks ( ) [override], [default]
```

Destructor.

8.2.3 Member Function Documentation

8.2.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassBodyLinks::~JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassBodyLinks ) [private]
```

8.2.3.2 operator=()

```
void jeod::MassBodyLinks::operator= (
    const MassBodyLinks & ) [delete]
```

8.2.4 Field Documentation

8.2.4.1 default_path_size

```
constexpr unsigned int jeod::MassBodyLinks::default_path_size {8} [static], [constexpr],
[private]
```

Definition at line 106 of file mass_body_links.hh.

The documentation for this class was generated from the following file:

- [mass_body_links.hh](#)

8.3 jeod::MassBodyMessages Class Reference

Specify the message IDs used in the [MassBody](#) model.

```
#include <mass_messages.hh>
```

Public Member Functions

- [MassBodyMessages](#) ()=delete
- [MassBodyMessages](#) (const [MassBodyMessages](#) &)=delete
- [MassBodyMessages](#) & operator= (const [MassBodyMessages](#) &)=delete

Static Public Attributes

- static const char * [attach_info](#) = "dynamics/mass/" "attach_info"
Issued to provide information regarding an attachment.
- static const char * [invalid_attach](#) = "dynamics/mass/" "invalid_attach"
Issued when an attachment cannot be performed as requested.
- static const char * [invalid_detach](#) = "dynamics/mass/" "invalid_detach"
Issued when a detachment cannot be performed as requested.
- static const char * [invalid_node](#) = "dynamics/mass/" "invalid_node"
Issued when a node does not have expected linkages.
- static const char * [invalid_name](#) = "dynamics/mass/" "invalid_name"
Issued when a name is invalid – NULL, empty, a duplicate, ...
- static const char * [invalid_enum](#) = "dynamics/mass/" "invalid_enum"
Issued when a enum value is not one of the enumerated values.
- static const char * [io_error](#) = "dynamics/mass/" "io_error"
Issued when an I/O error occurs.
- static const char * [internal_error](#) = "dynamics/mass/" "internal_error"
Error issued when some internal error occurred.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [MassBodyMessages](#))

8.3.1 Detailed Description

Specify the message IDs used in the [MassBody](#) model.

Assumptions and Limitations

- This is a complete catalog of all the messages sent by the [MassBody](#) model.
- This is not an exhaustive list of all the things that can go awry.

Definition at line 81 of file mass_messages.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 MassBodyMessages() [1/2]

```
jeod::MassBodyMessages::MassBodyMessages ( ) [delete]
```

8.3.2.2 MassBodyMessages() [2/2]

```
jeod::MassBodyMessages::MassBodyMessages (
    const MassBodyMessages & ) [delete]
```

8.3.3 Member Function Documentation

8.3.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassBodyMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassBodyMessages ) [private]
```

8.3.3.2 operator=()

```
MassBodyMessages& jeod::MassBodyMessages::operator= (
    const MassBodyMessages & ) [delete]
```

8.3.4 Field Documentation

8.3.4.1 attach_info

```
char const * jeod::MassBodyMessages::attach_info = "dynamics/mass/" "attach_info" [static]
```

Issued to provide information regarding an attachment.

trick_units(-)

Definition at line 88 of file mass_messages.hh.

Referenced by jeod::MassBody::attach_to().

8.3.4.2 internal_error

```
char const * jeod::MassBodyMessages::internal_error = "dynamics/mass/" "internal_error" [static]
```

Error issued when some internal error occurred.

These errors should never happen. `trick_units(-)`

Definition at line 124 of file `mass_messages.hh`.

8.3.4.3 invalid_attach

```
char const * jeod::MassBodyMessages::invalid_attach = "dynamics/mass/" "invalid_attach" [static]
```

Issued when an attachment cannot be performed as requested.

`trick_units(-)`

Definition at line 93 of file `mass_messages.hh`.

Referenced by `jeod::MassBody::attach_root_body()`, `jeod::MassBody::attach_to()`, `jeod::MassBody::attach_validate_child()`, `jeod::MassBody::attach_validate_parent()`, and `jeod::generate_bad_point_message()`.

8.3.4.4 invalid_detach

```
char const * jeod::MassBodyMessages::invalid_detach = "dynamics/mass/" "invalid_detach" [static]
```

Issued when a detachment cannot be performed as requested.

`trick_units(-)`

Definition at line 98 of file `mass_messages.hh`.

Referenced by `jeod::MassBody::detach()`, `jeod::MassBody::detach_validate_child()`, `jeod::MassBody::detach_validate_parent()`, and `jeod::MassBody::~~MassBody()`.

8.3.4.5 invalid_enum

```
char const * jeod::MassBodyMessages::invalid_enum = "dynamics/mass/" "invalid_enum" [static]
```

Issued when a enum value is not one of the enumerated values.

`trick_units(-)`

Definition at line 113 of file `mass_messages.hh`.

Referenced by `jeod::MassPointInit::initialize_mass_point()`, and `jeod::MassPropertiesInit::initialize_mass_properties()`.

8.3.4.6 invalid_name

```
char const * jeod::MassBodyMessages::invalid_name = "dynamics/mass/" "invalid_name" [static]
```

Issued when a name is invalid – NULL, empty, a duplicate, ...

trick_units(–)

Definition at line 108 of file mass_messages.hh.

Referenced by jeod::MassBody::add_mass_point().

8.3.4.7 invalid_node

```
char const * jeod::MassBodyMessages::invalid_node = "dynamics/mass/" "invalid_node" [static]
```

Issued when a node does not have expected linkages.

trick_units(–)

Definition at line 103 of file mass_messages.hh.

Referenced by jeod::MassPoint::compute_pred_rel_state(), jeod::MassPoint::compute_relative_state(), jeod::↔
MassPoint::compute_state_wrt_pred(), and jeod::MassBody::reattach().

8.3.4.8 io_error

```
char const * jeod::MassBodyMessages::io_error = "dynamics/mass/" "io_error" [static]
```

Issued when an I/O error occurs.

trick_units(–)

Definition at line 118 of file mass_messages.hh.

Referenced by jeod::MassBody::print_tree().

The documentation for this class was generated from the following files:

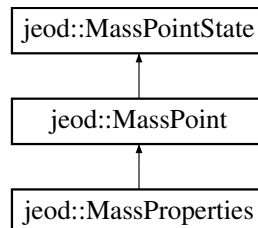
- [mass_messages.hh](#)
- [mass_messages.cc](#)

8.4 jeod::MassPoint Class Reference

Adds tree linkages and a name to a [MassPointState](#).

```
#include <mass_point.hh>
```

Inheritance diagram for jeod::MassPoint:



Public Member Functions

- [MassPoint](#) ()
Default constructor.
- [~MassPoint](#) () override
Destroy a [MassPoint](#) object.
- [MassPoint](#) (const [MassPoint](#) &)=delete
- [MassPoint](#) & operator= (const [MassPoint](#) &)=delete
- void [initialize_mass_point](#) () override
Initialize a mass point.
- void [set_name](#) (std::string name_in)
- std::string [get_name](#) () const
- virtual void [compute_relative_state](#) (const [MassPoint](#) &ref_point, [MassPointState](#) &rel_state) const
Compute the state of this point with respect to some reference point, which may be any point in the same tree as this point.
- virtual void [compute_state_wrt_pred](#) (const [MassPoint](#) &ref_point, [MassPointState](#) &rel_state) const
Compute the state of this point with respect to some reference point, which must be at or above this point via the parent links.
- virtual void [compute_state_wrt_pred](#) (unsigned int ref_point_index, [MassPointState](#) &rel_state) const
Compute the state of this point with respect to some reference point, which must be at or above this point via the parent links.
- virtual void [compute_pred_rel_state](#) (const [MassPoint](#) &ref_point, [MassPointState](#) &rel_state) const
Compute the complete state of the invoking reference frame with respect to the supplied reference frame, which must be a predecessor of the invoking frame.
- virtual void [compute_pred_rel_state](#) (unsigned int ref_point_index, [MassPointState](#) &rel_state) const
Compute the state of some reference point with respect to this point.
- const [MassPoint](#) * [find_last_common_node](#) (const [MassPoint](#) &ref_point) const
Each mass point has a path from the root of the mass point tree to the point in question.

Protected Member Functions

- int [find_last_common_index](#) (const [MassPoint](#) &ref_point) const
Each mass point has a path from the root of the mass point tree to the point in question.

Protected Attributes

- NamedItem [name](#)
The name of the mass point.
- [MassPointLinks](#) [links](#)
Linkage to rest of mass tree.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) f (jeod, [MassPoint](#))
- void [attach](#) ([MassPoint](#) &parent)
Attach a mass point to another.
- void [detach](#) ()
Detach a mass point from its parent.

Friends

- class [MassPointLinks](#)
- class [MassBody](#)

Additional Inherited Members

8.4.1 Detailed Description

Adds tree linkages and a name to a [MassPointState](#).

Definition at line 86 of file mass_point.hh.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 [MassPoint\(\)](#) [1/2]

```
jeod::MassPoint::MassPoint ( ) [inline]
```

Default constructor.

Definition at line 114 of file mass_point.hh.

8.4.2.2 ~MassPoint()

```
jeod::MassPoint::~~MassPoint ( ) [override]
```

Destroy a [MassPoint](#) object.

Definition at line 58 of file mass_point.cc.

References links.

8.4.2.3 MassPoint() [2/2]

```
jeod::MassPoint::MassPoint (
    const MassPoint & ) [delete]
```

8.4.3 Member Function Documentation

8.4.3.1 attach()

```
void jeod::MassPoint::attach (
    MassPoint & parent ) [inline], [private]
```

Attach a mass point to another.

Assumptions and Limitations

- This method only addresses the linkages. Some external agent must address the physical relation.

Parameters

<i>in, out</i>	<i>parent</i>	parent node
----------------	---------------	-------------

Definition at line 226 of file mass_point.hh.

Referenced by `jeod::MassBody::add_mass_point()`, `jeod::MassBody::attach_establish_links()`, `jeod::MassBody::attach_to()`, and `jeod::MassBody::MassBody()`.

8.4.3.2 compute_pred_rel_state() [1/2]

```
void jeod::MassPoint::compute_pred_rel_state (
    const MassPoint & ref_point,
    MassPointState & rel_state ) const [virtual]
```

Compute the complete state of the invoking reference frame with respect to the supplied reference frame, which *must* be a predecessor of the invoking frame.

Assumptions and Limitations

- The predecessor frame is a predecessor.

Parameters

in	<i>ref_point</i>	The point with respect to which the state is to be expressed
out	<i>rel_state</i>	The relative state

Definition at line 231 of file mass_point.cc.

References jeod::MassBodyMessages::invalid_node, and links.

Referenced by compute_relative_state().

8.4.3.3 compute_pred_rel_state() [2/2]

```
void jeod::MassPoint::compute_pred_rel_state (
    unsigned int ref_point_index,
    MassPointState & rel_state ) const [virtual]
```

Compute the state of some reference point with respect to this point.

The reference point must be at or above this point via the parent links.

Parameters

in	<i>ref_point_index</i>	Reference point index
out	<i>rel_state</i>	Relative state

Definition at line 258 of file mass_point.cc.

References jeod::MassPointState::compute_transformation(), jeod::MassPointState::initialize_mass_point(), links, jeod::MassPointState::position, jeod::MassPointState::Q_parent_this, and jeod::MassPointState::T_parent_this.

8.4.3.4 compute_relative_state()

```
void jeod::MassPoint::compute_relative_state (
    const MassPoint & ref_point,
    MassPointState & rel_state ) const [virtual]
```

Compute the state of this point with respect to some reference point, which may be any point in the same tree as this point.

Parameters

in	<i>ref_point</i>	The point with respect to which the state is to be expressed
out	<i>rel_state</i>	The relative state

Definition at line 87 of file mass_point.cc.

References `compute_pred_rel_state()`, `compute_state_wrt_pred()`, `jeod::MassPointState::decr_left()`, `find_last_common_index()`, `jeod::MassPointState::initialize_mass_point()`, `jeod::MassBodyMessages::invalid_node`, and `links`.

8.4.3.5 compute_state_wrt_pred() [1/2]

```
void jeod::MassPoint::compute_state_wrt_pred (
    const MassPoint & ref_point,
    MassPointState & rel_state ) const [virtual]
```

Compute the state of this point with respect to some reference point, which must be at or above this point via the parent links.

Parameters

in	<i>ref_point</i>	Reference point
out	<i>rel_state</i>	Point state wrt ref. pt.

Definition at line 155 of file mass_point.cc.

References `jeod::MassBodyMessages::invalid_node`, and `links`.

Referenced by `jeod::MassBody::attach_to()`, and `compute_relative_state()`.

8.4.3.6 compute_state_wrt_pred() [2/2]

```
void jeod::MassPoint::compute_state_wrt_pred (
    unsigned int ref_point_index,
    MassPointState & rel_state ) const [virtual]
```

Compute the state of this point with respect to some reference point, which must be at or above this point via the parent links.

Parameters

in	<i>ref_point_index</i>	Reference point index
out	<i>rel_state</i>	Point state wrt ref. pt.

Definition at line 181 of file mass_point.cc.

References `jeod::MassPointState::compute_transformation()`, `jeod::MassPointState::initialize_mass_point()`, `links`, `jeod::MassPointState::position`, `jeod::MassPointState::Q_parent_this`, and `jeod::MassPointState::T_parent_this`.

8.4.3.7 detach()

```
void jeod::MassPoint::detach ( ) [inline], [private]
```

Detach a mass point from its parent.

Assumptions and Limitations

- This method only addresses the linkages. Some external agent must address the physical relation.

Definition at line 238 of file mass_point.hh.

Referenced by jeod::MassBody::attach_to(), initialize_mass_point(), and jeod::MassBody::~~MassBody().

8.4.3.8 f()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::MassPoint::f (
    jeod ,
    MassPoint ) [private]
```

8.4.3.9 find_last_common_index()

```
int jeod::MassPoint::find_last_common_index (
    const MassPoint & ref_point ) const [inline], [protected]
```

Each mass point has a path from the root of the mass point tree to the point in question.

The paths for two mass points will have some initial sequence of common nodes. Find the index number of this last element in this sequence.

Returns

Last common node

Parameters

in	<i>ref_point</i>	Other point
----	------------------	-------------

Definition at line 192 of file mass_point.hh.

Referenced by compute_relative_state().

8.4.3.10 find_last_common_node()

```
const MassPoint * jeod::MassPoint::find_last_common_node (
    const MassPoint & frame ) const [inline]
```

Each mass point has a path from the root of the mass point tree to the point in question.

The paths for two mass points will have some initial sequence of common nodes. Find the last element in this sequence.

Returns

Last common node

Parameters

<code>in</code>	<code>frame</code>	Other point
-----------------	--------------------	-------------

Definition at line 205 of file mass_point.hh.

8.4.3.11 get_name()

```
std::string jeod::MassPoint::get_name ( ) const [inline]
```

Definition at line 135 of file mass_point.hh.

References [name](#).

8.4.3.12 initialize_mass_point()

```
void jeod::MassPoint::initialize_mass_point ( ) [override], [virtual]
```

Initialize a mass point.

Reimplemented from [jeod::MassPointState](#).

Definition at line 75 of file mass_point.cc.

References [detach\(\)](#), and [jeod::MassPointState::initialize_mass_point\(\)](#).

Referenced by [jeod::MassBody::detach_update_properties\(\)](#).

8.4.3.13 operator=()

```
MassPoint& jeod::MassPoint::operator= (
    const MassPoint & ) [delete]
```

8.4.3.14 set_name()

```
void jeod::MassPoint::set_name (
    std::string name_in ) [inline]
```

Definition at line 129 of file mass_point.hh.

References name.

8.4.4 Friends And Related Function Documentation

8.4.4.1 MassBody

```
friend class MassBody [friend]
```

Definition at line 89 of file mass_point.hh.

8.4.4.2 MassPointLinks

```
friend class MassPointLinks [friend]
```

Definition at line 88 of file mass_point.hh.

8.4.5 Field Documentation

8.4.5.1 links

`MassPointLinks` jeod::MassPoint::links [protected]

Linkage to rest of mass tree.

Programmatic interfaces:

- `MassPointLinks` provides accessors to the parent and root and provides methods to attach, detach links (and hence bodies).
- This class provides accessors to the same.
- Various iterators provide the ability to iterate over child bodies and up the parent chain. `trick_units(-)`

Definition at line 107 of file `mass_point.hh`.

Referenced by `compute_pred_rel_state()`, `compute_relative_state()`, `compute_state_wrt_pred()`, and `~MassPoint()`.

8.4.5.2 name

`NamedItem` jeod::MassPoint::name [protected]

The name of the mass point.

`trick_units(-)`

Definition at line 96 of file `mass_point.hh`.

Referenced by `jeod::MassBody::add_mass_point()`, `get_name()`, and `set_name()`.

The documentation for this class was generated from the following files:

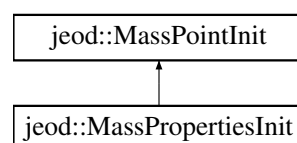
- `mass_point.hh`
- `mass_point.cc`

8.5 jeod::MassPointInit Class Reference

Contains data used to initialize a `MassPoint`.

```
#include <mass_point_init.hh>
```

Inheritance diagram for `jeod::MassPointInit`:



Public Types

- enum `FrameSpec` {
`StructToBody` = 0 , `StructToCase` = 0 , `StructToPoint` = 0 , `StructToChild` = 0 ,
`BodyToStruct` = 1 , `CaseToStruct` = 1 , `PointToStruct` = 1 , `ChildToStruct` = 1 }
Specifies sense of the input point orientation data.

Public Member Functions

- `MassPointInit` ()
- virtual `~MassPointInit` ()=default
- void `initialize_mass_point` (`MassPoint` &mass_point) const
Default constructor; constructs a `MassPointInit` object.
- void `set_name` (std::string name_in)
Set the name.

Data Fields

- double `position` [3] {}
Mass point location expressed in mass element structural coordinates.
- Orientation `pt_orientation`
Mass point frame orientation specification.
- `FrameSpec` `pt_frame_spec` {`StructToPoint`}
Indicates whether user orientation input defines the structure-to-body or body-to-structure transformation matrix.
- std::string `name` {""}
Item name, with the following semantics for items that have a name:

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` `p` (`jeod`, `MassPointInit`)

8.5.1 Detailed Description

Contains data used to initialize a `MassPoint`.

Definition at line 87 of file `mass_point_init.hh`.

8.5.2 Member Enumeration Documentation

8.5.2.1 `FrameSpec`

```
enum jeod::MassPointInit::FrameSpec
```

Specifies sense of the input point orientation data.

The overloading of names is intentional.

Enumerator

StructToBody	Orientation specifies structure-to-body transform.
StructToCase	Orientation specifies structure-to-case transform.
StructToPoint	Orientation specifies structure-to-point transform.
StructToChild	Orientation specifies structure-to-child transform.
BodyToStruct	Orientation specifies body-to-structure transform.
CaseToStruct	Orientation specifies case-to-structure transform.
PointToStruct	Orientation specifies point-to-structure transform.
ChildToStruct	Orientation specifies child-to-structure transform.

Definition at line 94 of file mass_point_init.hh.

8.5.3 Constructor & Destructor Documentation

8.5.3.1 MassPointInit()

```
jeod::MassPointInit::MassPointInit ( )
```

Definition at line 51 of file mass_point_init.cc.

8.5.3.2 ~MassPointInit()

```
virtual jeod::MassPointInit::~~MassPointInit ( ) [virtual], [default]
```

8.5.4 Member Function Documentation

8.5.4.1 initialize_mass_point()

```
void jeod::MassPointInit::initialize_mass_point (
    MassPoint & mass_point ) const
```

Default constructor; constructs a [MassPointInit](#) object.

Parameters

out	<i>mass_point</i>	Point to initialize
-----	-------------------	---------------------

Definition at line 60 of file mass_point_init.cc.

References jeod::MassBodyMessages::invalid_enum, PointToStruct, position, pt_frame_spec, pt_orientation, jeod::MassPointState::Q_parent_this, StructToPoint, jeod::MassPointState::T_parent_this, and jeod::MassPointState::update_point().

Referenced by jeod::MassBody::add_mass_point(), and jeod::MassPropertiesInit::initialize_mass_properties().

8.5.4.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::MassPointInit::p (
    jeod ,
    MassPointInit ) [private]
```

8.5.4.3 set_name()

```
void jeod::MassPointInit::set_name (
    std::string name_in ) [inline]
```

Set the name.

Definition at line 144 of file mass_point_init.hh.

8.5.5 Field Documentation

8.5.5.1 name

```
std::string jeod::MassPointInit::name {""}
```

Item name, with the following semantics for items that have a name:

- The mass point name will always be of the form "mass_name.point_name".
- If the supplied name does not begin with "mass_name.", this prefix will be applied to the supplied name in naming the mass point.

This can be left as the empty string for items that don't have a name. trick_units(-)

Definition at line 132 of file mass_point_init.hh.

Referenced by jeod::MassBody::add_mass_point().

8.5.5.2 position

```
double jeod::MassPointInit::position[3] {}
```

Mass point location expressed in mass element structural coordinates.

trick_units(m)

Definition at line 111 of file mass_point_init.hh.

Referenced by initialize_mass_point(), and jeod::MassPropertiesInit::MassPropertiesInit().

8.5.5.3 pt_frame_spec

```
FrameSpec jeod::MassPointInit::pt_frame_spec {StructToPoint}
```

Indicates whether user orientation input defines the structure-to-body or body-to-structure transformation matrix.

trick_units(-)

Definition at line 122 of file mass_point_init.hh.

Referenced by initialize_mass_point().

8.5.5.4 pt_orientation

```
Orientation jeod::MassPointInit::pt_orientation
```

Mass point frame orientation specification.

trick_units(-)

Definition at line 116 of file mass_point_init.hh.

Referenced by initialize_mass_point().

The documentation for this class was generated from the following files:

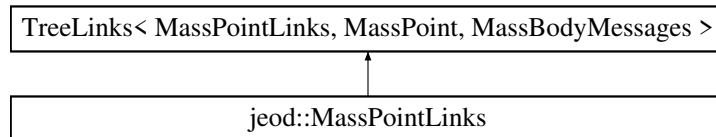
- [mass_point_init.hh](#)
- [mass_point_init.cc](#)

8.6 jeod::MassPointLinks Class Reference

Encapsulates the links between mass points.

```
#include <mass_point_links.hh>
```

Inheritance diagram for jeod::MassPointLinks:



Public Member Functions

- [MassPointLinks](#) ([MassPoint](#) &container_in)
Non-default constructor.
- [MassPointLinks](#) ()=delete
- [MassPointLinks](#) (const [MassPointLinks](#) &)=delete
- void [operator=](#) (const [MassPointLinks](#) &)=delete
- [~MassPointLinks](#) () override=default
Destructor.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [MassPointLinks](#))

Static Private Attributes

- static constexpr unsigned int [default_path_size](#) {8}

8.6.1 Detailed Description

Encapsulates the links between mass points.

Definition at line 81 of file mass_point_links.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 MassPointLinks() [1/3]

```
jeod::MassPointLinks::MassPointLinks (
    MassPoint & container_in ) [inline], [explicit]
```

Non-default constructor.

Parameters

<i>container</i> <i>_in</i>	The MassPoint object that contains this object.
--------------------------------	---

Definition at line 89 of file mass_point_links.hh.

8.6.2.2 MassPointLinks() [2/3]

```
jeod::MassPointLinks::MassPointLinks ( ) [delete]
```

8.6.2.3 MassPointLinks() [3/3]

```
jeod::MassPointLinks::MassPointLinks (
    const MassPointLinks & ) [delete]
```

8.6.2.4 ~MassPointLinks()

```
jeod::MassPointLinks::~~MassPointLinks ( ) [override], [default]
```

Destructor.

8.6.3 Member Function Documentation

8.6.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassPointLinks::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassPointLinks ) [private]
```

8.6.3.2 operator=()

```
void jeod::MassPointLinks::operator= (
    const MassPointLinks & ) [delete]
```

8.6.4 Field Documentation

8.6.4.1 default_path_size

```
constexpr unsigned int jeod::MassPointLinks::default_path_size {8} [static], [constexpr],
[private]
```

Definition at line 106 of file mass_point_links.hh.

The documentation for this class was generated from the following file:

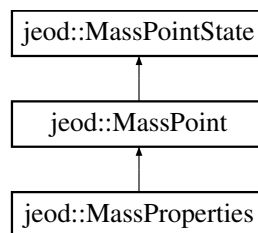
- [mass_point_links.hh](#)

8.7 jeod::MassPointState Class Reference

Defines the state – position and orientation – of a [MassPoint](#).

```
#include <mass_point_state.hh>
```

Inheritance diagram for jeod::MassPointState:



Public Member Functions

- [MassPointState](#) ()=default
- virtual [~MassPointState](#) ()=default
- virtual void [initialize_mass_point](#) ()
Initialize a mass point.
- void [update_point](#) (const double pt_location[3])
Update the mass point's structure-to-point vector.
- void [update_orientation](#) (const double transformation[3][3])
Update the mass point's structure-to-point frame transformation matrix and items derived from it.
- void [update_orientation](#) (const Quaternion &left_quat)
Update the mass point's structure-to-point frame transformation matrix and items derived from it.
- void [compute_quaternion](#) ()
Compute quaternion.
- void [compute_transformation](#) ()
Compute matrix.
- void [copy_state](#) (const [MassPointState](#) &source)

- *Copy the provided state to this.*
- void [negate](#) (const [MassPointState](#) &source)
- *Copy a mass point state, negated.*
- void [incr_left](#) (const [MassPointState](#) &s_ab)
- *Compute $S_A:C = S_A:B + S_B:C$, with this initially containing $S_B:C$, the supplied argument containing $S_A:B$, and the resultant composition of states stored in this.*
- void [incr_right](#) (const [MassPointState](#) &s_bc)
- *Compute $S_A:C = S_A:B + S_B:C$, with this initially containing $S_A:B$, the supplied argument containing $S_B:C$, and the resultant composition of states stored in this.*
- void [decr_left](#) (const [MassPointState](#) &s_ab)
- *Compute $S_B:C = (-S_A:B) + S_A:C$, with this initially containing $S_A:C$, the supplied argument containing $S_A:B$, and the resultant composition of states stored in this.*
- void [decr_right](#) (const [MassPointState](#) &s_bc)
- *Compute $S_A:B = S_A:C + (-S_B:C)$ with this initially containing $S_A:C$, the supplied argument containing $S_B:C$, and the resultant composition of states stored in this.*

Data Fields

- double [position](#) [3] {}
- *Mass point location with respect to the origin of some parent frame and expressed in the parent frame's coordinates.*
- Quaternion [Q_parent_this](#)
- *Left transformation quaternion from the parent frame to the frame associated with the mass point.*
- double [T_parent_this](#) [3][3] { { 1.0, 0.0, 0.0 }, { 0.0, 1.0, 0.0 }, { 0.0, 0.0, 1.0 } }
- *transformation matrix from the parent frame to the frame associated with the mass point.*

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [MassPointState](#))

8.7.1 Detailed Description

Defines the state – position and orientation – of a [MassPoint](#).

Definition at line 84 of file `mass_point_state.hh`.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 MassPointState()

```
jeod::MassPointState::MassPointState ( ) [default]
```

8.7.2.2 ~MassPointState()

```
virtual jeod::MassPointState::~~MassPointState ( ) [virtual], [default]
```

8.7.3 Member Function Documentation

8.7.3.1 compute_quaternion()

```
void jeod::MassPointState::compute_quaternion ( ) [inline]
```

Compute quaternion.

Someone changed the matrix under our noses.

Definition at line 191 of file mass_point_state.hh.

8.7.3.2 compute_transformation()

```
void jeod::MassPointState::compute_transformation ( ) [inline]
```

Compute matrix.

Someone changed the quaternion under our noses.

Definition at line 199 of file mass_point_state.hh.

Referenced by jeod::MassBody::attach_update_properties(), jeod::MassPoint::compute_pred_rel_state(), jeod::MassPoint::compute_state_wrt_pred(), decr_left(), decr_right(), incr_left(), incr_right(), and jeod::MassBody::reattach().

8.7.3.3 copy_state()

```
void jeod::MassPointState::copy_state (
    const MassPointState & source ) [inline]
```

Copy the provided state to this.

Parameters

in	<i>source</i>	Source state
----	---------------	--------------

Definition at line 209 of file mass_point_state.hh.

8.7.3.4 decr_left()

```
void jeod::MassPointState::decr_left (
    const MassPointState & s_ab )
```

Compute $S_{B:C} = (-S_{A:B}) + S_{A:C}$, with this initially containing $S_{A:C}$, the supplied argument containing $S_{A:B}$, and the resultant composition of states stored in this.

Parameters

in	s_ab	Left addend
----	------	-------------

Definition at line 240 of file mass_point_state.cc.

References compute_transformation(), position, Q_parent_this, and T_parent_this.

Referenced by jeod::MassPoint::compute_relative_state().

8.7.3.5 decr_right()

```
void jeod::MassPointState::decr_right (
    const MassPointState & s_bc )
```

Compute $S_{A:B} = S_{A:C} + (-S_{B:C})$ with this initially containing $S_{A:C}$, the supplied argument containing $S_{B:C}$, and the resultant composition of states stored in this.

Parameters

in	s_bc	Left addend
----	------	-------------

Definition at line 283 of file mass_point_state.cc.

References compute_transformation(), position, Q_parent_this, and T_parent_this.

8.7.3.6 incr_left()

```
void jeod::MassPointState::incr_left (
    const MassPointState & s_ab )
```

Compute $S_{A:C} = S_{A:B} + S_{B:C}$, with this initially containing $S_{B:C}$, the supplied argument containing $S_{A:B}$, and the resultant composition of states stored in this.

Parameters

in	<i>s_ab</i>	Left addend
----	-------------	-------------

Definition at line 154 of file mass_point_state.cc.

References `compute_transformation()`, `position`, `Q_parent_this`, and `T_parent_this`.

8.7.3.7 incr_right()

```
void jeod::MassPointState::incr_right (
    const MassPointState & s_bc )
```

Compute $S_{A:C} = S_{A:B} + S_{B:C}$, with this initially containing $S_{A:B}$, the supplied argument containing $S_{B:C}$, and the resultant composition of states stored in this.

Parameters

in	<i>s_bc</i>	Right addend
----	-------------	--------------

Definition at line 195 of file mass_point_state.cc.

References `compute_transformation()`, `position`, `Q_parent_this`, and `T_parent_this`.

8.7.3.8 initialize_mass_point()

```
void jeod::MassPointState::initialize_mass_point ( ) [virtual]
```

Initialize a mass point.

Reimplemented in [jeod::MassPoint](#).

Definition at line 52 of file mass_point_state.cc.

References `position`, `Q_parent_this`, and `T_parent_this`.

Referenced by `jeod::MassPoint::compute_pred_rel_state()`, `jeod::MassPoint::compute_relative_state()`, `jeod::↔
MassPoint::compute_state_wrt_pred()`, and `jeod::MassPoint::initialize_mass_point()`.

8.7.3.9 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassPointState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassPointState ) [private]
```

8.7.3.10 negate()

```
void jeod::MassPointState::negate (
    const MassPointState & source )
```

Copy a mass point state, negated.

Parameters

in	<i>source</i>	Source state
----	---------------	--------------

Definition at line 120 of file `mass_point_state.cc`.

References `position`, `Q_parent_this`, and `T_parent_this`.

8.7.3.11 update_orientation() [1/2]

```
void jeod::MassPointState::update_orientation (
    const double transformation[3][3] ) [inline]
```

Update the mass point's structure-to-point frame transformation matrix and items derived from it.

Parameters

in	<i>transformation</i>	New structure-to-point xform
----	-----------------------	------------------------------

Definition at line 169 of file `mass_point_state.hh`.

Referenced by `jeod::MassBody::attach_update_properties()`, and `jeod::MassBody::reattach()`.

8.7.3.12 update_orientation() [2/2]

```
void jeod::MassPointState::update_orientation (
    const Quaternion & left_quat ) [inline]
```

Update the mass point's structure-to-point frame transformation matrix and items derived from it.

Parameters

in	<i>left_quat</i>	New structure-to-point quat
----	------------------	-----------------------------

Definition at line 181 of file `mass_point_state.hh`.

8.7.3.13 update_point()

```
void jeod::MassPointState::update_point (
    const double pt_location[3] ) [inline]
```

Update the mass point's structure-to-point vector.

Parameters

in	<i>pt_location</i>	New location, struct coords. Units: M
----	--------------------	--

Definition at line 158 of file mass_point_state.hh.

Referenced by `jeod::MassBody::attach_update_properties()`, `jeod::MassPointInit::initialize_mass_point()`, and `jeod::MassBody::reattach()`.

8.7.4 Field Documentation

8.7.4.1 position

```
double jeod::MassPointState::position[3] {}
```

Mass point location with respect to the origin of some parent frame and expressed in the parent frame's coordinates.

`trick_units(m)`

Definition at line 92 of file mass_point_state.hh.

Referenced by `jeod::MassBody::attach_to()`, `jeod::MassBody::attach_update_properties()`, `jeod::MassBody::calc_composite_cm()`, `jeod::MassBody::calc_composite_inertia()`, `jeod::MassPoint::compute_pred_rel_state()`, `jeod::MassPoint::compute_state_wrt_pred()`, `decr_left()`, `decr_right()`, `incr_left()`, `incr_right()`, `initialize_mass_point()`, `jeod::MassPropertiesInit::initialize_mass_properties()`, `negate()`, `jeod::MassBody::print_body()`, `jeod::MassBody::reattach()`, and `jeod::MassBody::update_mass_properties()`.

8.7.4.2 Q_parent_this

```
Quaternion jeod::MassPointState::Q_parent_this
```

Left transformation quaternion from the parent frame to the frame associated with the mass point.

`trick_units(-)`

Definition at line 98 of file mass_point_state.hh.

Referenced by `jeod::MassBody::attach_to()`, `jeod::MassBody::attach_update_properties()`, `jeod::MassPoint::compute_pred_rel_state()`, `jeod::MassPoint::compute_state_wrt_pred()`, `decr_left()`, `decr_right()`, `incr_left()`, `incr_right()`, `jeod::MassBody::initialize_mass()`, `initialize_mass_point()`, `jeod::MassPointInit::initialize_mass_point()`, `negate()`, and `jeod::MassBody::reattach()`.

8.7.4.3 T_parent_this

```
double jeod::MassPointState::T_parent_this[3][3] { {1.0, 0.0, 0.0}, {0.0, 1.0, 0.0}, {0.0, 0.0, 1.0} }
```

transformation matrix from the parent frame to the frame associated with the mass point.

NOTE: The [MassPointState](#) infrastructure ensures that the quaternion and matrix representations are equivalent. `trick_units(-)`

Definition at line 106 of file `mass_point_state.hh`.

Referenced by `jeod::MassBody::attach_to()`, `jeod::MassBody::attach_update_properties()`, `jeod::MassBody::calc_composite_inertia()`, `jeod::MassPoint::compute_pred_rel_state()`, `jeod::MassPoint::compute_state_wrt_pred()`, `decr_left()`, `decr_right()`, `incr_left()`, `incr_right()`, `jeod::MassBody::initialize_mass()`, `initialize_mass_point()`, `jeod::MassPointInit::initialize_mass_point()`, `jeod::MassPropertiesInit::initialize_mass_properties()`, `jeod::MassBody::MassBody()`, `negate()`, `jeod::MassBody::print_body()`, `jeod::MassBody::reattach()`, and `jeod::MassBody::update_mass_properties()`.

The documentation for this class was generated from the following files:

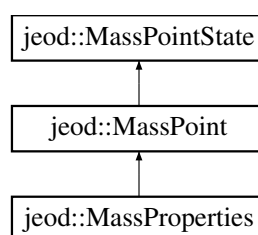
- [mass_point_state.hh](#)
- [mass_point_state.cc](#)

8.8 jeod::MassProperties Class Reference

Defines mass properties – mass and inertia tensor.

```
#include <mass_properties.hh>
```

Inheritance diagram for `jeod::MassProperties`:



Public Member Functions

- [MassProperties](#) ()=default
- [MassProperties](#) (const [MassProperties](#) &)=delete
- [MassProperties](#) & operator= (const [MassProperties](#) &)=delete

Data Fields

- double `mass` {}
Mass of the subject mass element (always ≥ 0).
- double `inertia` [3][3] {}
Inertia tensor of the subject mass element about the subject mass element's center of mass, expressed in subject mass element body coordinates.
- double `inverse_mass` {}
The inverse of mass.
- double `inverse_inertia` [3][3] {}
The inverse of the composite inertia.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` (jeod, `MassProperties`)

Additional Inherited Members

8.8.1 Detailed Description

Defines mass properties – mass and inertia tensor.

A `MassBody` contains two `MassProperties` objects that represent the body's core and composite mass properties. A `MassProperties` object is-a `MassPoint` and hence a `MassPointState` by inheritance. The parent point of a `MassProperties` object is always the structural frame of the `MassBody` that 'owns' the `MassProperties` object.

Note that a `MassProperties` does not explicitly define the center of mass location and body frame orientation. These properties are defined by elements inherited from `MassPointState`:

- position defines the center of mass in the structural frame.
- `Q_parent_this/T_parent_this` define the orientation of the body frame with respect to the structural frame.

Definition at line 86 of file `mass_properties.hh`.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 `MassProperties()` [1/2]

```
jeod::MassProperties::MassProperties ( ) [default]
```

8.8.2.2 MassProperties() [2/2]

```
jeod::MassProperties::MassProperties (
    const MassProperties & ) [delete]
```

8.8.3 Member Function Documentation

8.8.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassProperties::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassProperties ) [private]
```

8.8.3.2 operator=()

```
MassProperties& jeod::MassProperties::operator= (
    const MassProperties & ) [delete]
```

8.8.4 Field Documentation

8.8.4.1 inertia

```
double jeod::MassProperties::inertia[3][3] {}
```

Inertia tensor of the subject mass element about the subject mass element's center of mass, expressed in subject mass element body coordinates.

Note: The diagonal elements of the inertia tensor are positive moments of inertial while the off-diagonal elements are megative products of inertia. trick_units(kg*m2)

Definition at line 104 of file mass_properties.hh.

Referenced by jeod::MassBody::calc_composite_inertia(), jeod::MassBody::detach_update_properties(), jeod↔
::MassPropertiesInit::initialize_mass_properties(), jeod::MassBody::print_body(), and jeod::MassBody::update_↔
mass_properties().

8.8.4.2 inverse_inertia

```
double jeod::MassProperties::inverse_inertia[3][3] {}
```

The inverse of the composite inertia.

Used only for root dynamic bodies composite props. trick_units(1/kg/m2)

Definition at line 116 of file mass_properties.hh.

Referenced by jeod::MassBody::detach_update_properties(), jeod::MassBody::print_body(), and jeod::MassBody::update_mass_properties().

8.8.4.3 inverse_mass

```
double jeod::MassProperties::inverse_mass {}
```

The inverse of mass.

Used only for root dynamic bodies composite props. trick_units(1/kg)

Definition at line 110 of file mass_properties.hh.

Referenced by jeod::MassBody::calc_composite_cm(), jeod::MassBody::print_body(), and jeod::MassBody::update_mass_properties().

8.8.4.4 mass

```
double jeod::MassProperties::mass {}
```

Mass of the subject mass element (always ≥ 0).

The core mass element's mass is set externally while the composite mass element's mass is computed by the mass model. trick_units(kg)

Definition at line 95 of file mass_properties.hh.

Referenced by jeod::MassBody::calc_composite_cm(), jeod::MassBody::calc_composite_inertia(), jeod::MassBody::detach_update_properties(), jeod::MassPropertiesInit::initialize_mass_properties(), jeod::MassBody::print_body(), and jeod::MassBody::update_mass_properties().

The documentation for this class was generated from the following file:

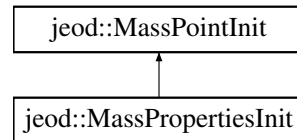
- [mass_properties.hh](#)

8.9 jeod::MassPropertiesInit Class Reference

Contains data used to initialize a mass model object.

```
#include <mass_properties_init.hh>
```

Inheritance diagram for jeod::MassPropertiesInit:



Public Types

- enum [InertiaSpec](#) {
[NoSpec](#) = 0 , [Body](#) = 1 , [StructCG](#) = 2 , [Struct](#) = 3 ,
[SpecCG](#) = 4 , [Spec](#) = 5 }
Specifies the origin and axes of the input inertia tensor.

Public Member Functions

- [MassPropertiesInit](#) ()
Default constructor; constructs a [MassPropertiesInit](#) object.
- void [initialize_mass_properties](#) ([MassProperties](#) &mass_properties) const
Initialize the subject [MassProperties](#).

Data Fields

- double [mass](#) {}
Mass of the core body.
- double * [cm](#)
Location of the core body center of mass in the structural frame.
- double [inertia](#) [3][3] {}
Inertia tensor of the core body in an arbitrary reference frame.
- [InertiaSpec](#) [inertia_spec](#) {[Body](#)}
Indicates how the user has specified the core inertia matrix.
- double [inertia_offset](#) [3] {}
Offset from inertia reference frame to the core center of mass, expressed in inertia reference frame coordinates.
- Orientation [inertia_orientation](#)
Orientation of body frame wrt inertia reference frame, i.e.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [MassPropertiesInit](#))

8.9.1 Detailed Description

Contains data used to initialize a mass model object.

Definition at line 79 of file mass_properties_init.hh.

8.9.2 Member Enumeration Documentation

8.9.2.1 InertiaSpec

```
enum jeod::MassPropertiesInit::InertiaSpec
```

Specifies the origin and axes of the input inertia tensor.

Enumerator

NoSpec	Inertia matrix is specified directly.
Body	Initial inertia frame is body frame.
StructCG	Initial inertia frame is struct at CG.
Struct	Initial inertia frame is structural frame.
SpecCG	Initial inertia frame is in frame at CG.
Spec	Initial inertia frame is specified frame.

Definition at line 87 of file mass_properties_init.hh.

8.9.3 Constructor & Destructor Documentation

8.9.3.1 MassPropertiesInit()

```
jeod::MassPropertiesInit::MassPropertiesInit ( )
```

Default constructor; constructs a [MassPropertiesInit](#) object.

Definition at line 55 of file mass_properties_init.cc.

References [cm](#), and [jeod::MassPointInit::position](#).

8.9.4 Member Function Documentation

8.9.4.1 initialize_mass_properties()

```
void jeod::MassPropertiesInit::initialize_mass_properties (
    MassProperties & properties ) const
```

Initialize the subject [MassProperties](#).

Parameters

out	<i>properties</i>	Properties to initialize
-----	-------------------	--------------------------

Definition at line 65 of file mass_properties_init.cc.

References Body, jeod::MassBody::compute_point_mass_inertia(), jeod::MassProperties::inertia, inertia, inertia↔_offset, inertia_orientation, inertia_spec, jeod::MassPointInit::initialize_mass_point(), jeod::MassBodyMessages↔::invalid_enum, jeod::MassProperties::mass, mass, NoSpec, jeod::MassPointState::position, Spec, SpecCG, Struct, StructCG, and jeod::MassPointState::T_parent_this.

Referenced by jeod::MassBody::initialize_mass().

8.9.4.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassPropertiesInit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassPropertiesInit ) [private]
```

8.9.5 Field Documentation

8.9.5.1 cm

```
double* jeod::MassPropertiesInit::cm
```

Location of the core body center of mass in the structural frame.

This is just an alias for the generic position member. trick_units(m)

Definition at line 108 of file mass_properties_init.hh.

Referenced by MassPropertiesInit().

8.9.5.2 inertia

```
double jeod::MassPropertiesInit::inertia[3][3] {}
```

Inertia tensor of the core body in an arbitrary reference frame.

trick_units(kg*m2)

Definition at line 113 of file mass_properties_init.hh.

Referenced by initialize_mass_properties().

8.9.5.3 inertia_offset

```
double jeod::MassPropertiesInit::inertia_offset[3] {}
```

Offset from inertia reference frame to the core center of mass, expressed in inertia reference frame coordinates.

trick_units(m)

Definition at line 124 of file mass_properties_init.hh.

Referenced by initialize_mass_properties().

8.9.5.4 inertia_orientation

```
Orientation jeod::MassPropertiesInit::inertia_orientation
```

Orientation of body frame wrt inertia reference frame, i.e.

the process by which the frame in which the inertia tensor is specified may be transformed to the body frame, e.g. $T_{\{\text{spec} \rightarrow \text{body}\}}$. trick_units(-)

Definition at line 131 of file mass_properties_init.hh.

Referenced by initialize_mass_properties().

8.9.5.5 inertia_spec

```
InertiaSpec jeod::MassPropertiesInit::inertia_spec {Body}
```

Indicates how the user has specified the core inertia matrix.

trick_units(-)

Definition at line 118 of file mass_properties_init.hh.

Referenced by initialize_mass_properties().

8.9.5.6 mass

```
double jeod::MassPropertiesInit::mass {}
```

Mass of the core body.

trick_units(kg)

Definition at line 102 of file mass_properties_init.hh.

Referenced by initialize_mass_properties().

The documentation for this class was generated from the following files:

- [mass_properties_init.hh](#)
- [mass_properties_init.cc](#)

Chapter 9

File Documentation

9.1 `class_declarations.hh` File Reference

Forward declarations of mass model classes.

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Forward declarations of mass model classes.

9.2 `mass.cc` File Reference

Define methods for the MassBody class.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
#include "../include/mass_properties_init.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.2.1 Detailed Description

Define methods for the MassBody class.

9.3 mass.hh File Reference

Define the class MassBody.

```
#include <list>
#include "class_declarations.hh"
#include "mass_body_links.hh"
#include "mass_point.hh"
#include "mass_point_init.hh"
#include "mass_properties.hh"
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "utils/container/include/pointer_list.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include <cstddef>
#include <cstdio>
#include <string>
#include <utility>
```

Data Structures

- class [jeod::MassBody](#)

Represent both an atomic chunk of mass and an interconnected set of masses.

Namespaces

- [jeod](#)

Namespace jeod.

9.3.1 Detailed Description

Define the class MassBody.

A MassBody represents both some atomic chunk of mass and an interconnected set of masses, and contains mass properties to represent both this atomic chunk of mass and the composite chunks of mass connected to this MassBody.

9.4 mass_attach.cc File Reference

Define MassBody attach methods.

```
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include <cstdint>
```

Namespaces

- [jeod](#)
Namespace jeod.

Functions

- static void [jeod::generate_bad_point_message](#) (const char *file, unsigned int line, const std::string &child_↵
body_name, const std::string &child_point_name, const MassPoint *child_point, const std::string &parent_↵
body_name, const std::string &parent_point_name, const MassPoint *parent_point)
Generate a message regarding failure to find mass points on a vehicle.

9.4.1 Detailed Description

Define MassBody attach methods.

9.5 mass_body_links.hh File Reference

Define the class MassBodyLinks, the class that encapsulates the links between mass bodies.

```
#include "utils/ref_frames/include/tree_links.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "mass_messages.hh"
```

Data Structures

- class [jeod::MassBodyLinks](#)
Encapsulates the links between mass bodies.

Namespaces

- [jeod](#)
Namespace jeod.

9.5.1 Detailed Description

Define the class MassBodyLinks, the class that encapsulates the links between mass bodies.

9.6 mass_calc_composite_cm.cc File Reference

Calculate the aggregate cm for a composite MassBody.

```
#include "utils/math/include/vector3.hh"
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include "../include/mass.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.6.1 Detailed Description

Calculate the aggregate cm for a composite MassBody.

9.7 mass_calc_composite_inertia.cc File Reference

Calculate the aggregate inertia matrix for a composite MassBody.

```
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include "../include/mass.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.7.1 Detailed Description

Calculate the aggregate inertia matrix for a composite MassBody.

9.8 mass_detach.cc File Reference

Define MassBody detachment methods.

```
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include <cstdlib>
```

Namespaces

- [jeod](#)
Namespace jeod.

9.8.1 Detailed Description

Define MassBody detachment methods.

9.9 mass_messages.cc File Reference

Implement the class MassBodyMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/mass_messages.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- `#define MAKE\_MASSBODY\_MESSAGE\_CODE(id) JEOD_MAKE_MESSAGE_CODE(MassBodyMessages, "dynamics/mass/", id)`

9.9.1 Detailed Description

Implement the class MassBodyMessages.

9.9.2 Macro Definition Documentation

9.9.2.1 MAKE_MASSBODY_MESSAGE_CODE

```
#define MAKE_MASSBODY_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(MassBodyMessages, "dynamics/mass/", id)
```

Definition at line 43 of file mass_messages.cc.

9.10 mass_messages.hh File Reference

Define the class MassBodyMessages, the class that specifies the message IDs used in the MassBody model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::MassBodyMessages](#)
Specify the message IDs used in the [MassBody](#) model.

Namespaces

- [jeod](#)
Namespace jeod.

9.10.1 Detailed Description

Define the class MassBodyMessages, the class that specifies the message IDs used in the MassBody model.

9.11 mass_point.cc File Reference

Define basic methods for the MassPoint class.

```
#include "../include/mass_point.hh"  
#include "../include/mass_messages.hh"  
#include "utils/math/include/matrix3x3.hh"  
#include "utils/math/include/vector3.hh"  
#include "utils/message/include/message_handler.hh"  
#include "utils/ref_frames/include/tree_links_iterator.hh"  
#include <cstdint>
```


Namespaces

- [jeod](#)

Namespace jeod.

9.11.1 Detailed Description

Define basic methods for the MassPoint class.

9.12 mass_point.hh File Reference

Define the class MassPoint, which defines the base features of a point related to a MassBody.

```
#include "class_declarations.hh"
#include "mass_point_links.hh"
#include "mass_point_state.hh"
#include "utils/named_item/include/named_item.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include <string>
#include <utility>
```

Data Structures

- class [jeod::MassPoint](#)

Adds tree linkages and a name to a [MassPointState](#).

Namespaces

- [jeod](#)

Namespace jeod.

9.12.1 Detailed Description

Define the class MassPoint, which defines the base features of a point related to a MassBody.

A MassPoint is simply a MassPointState with a name plus a pointer to a "parent" MassPoint.

9.13 mass_point_init.cc File Reference

Define methods for the MassPointInit class.

```
#include "../include/mass_point_init.hh"
#include "../include/mass_messages.hh"
#include "../include/mass_point.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.13.1 Detailed Description

Define methods for the MassPointInit class.

9.14 mass_point_init.hh File Reference

Define the class MassPointInit, which initializes a MassPoint object.

```
#include "class_declarations.hh"
#include "utils/orientation/include/orientation.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include <string>
#include <utility>
```

Data Structures

- class [jeod::MassPointInit](#)

Contains data used to initialize a [MassPoint](#).

Namespaces

- [jeod](#)

Namespace jeod.

9.14.1 Detailed Description

Define the class MassPointInit, which initializes a MassPoint object.

9.15 mass_point_links.hh File Reference

Define the class MassPointLinks, the class that encapsulates the links between mass points.

```
#include "class_declarations.hh"
#include "mass_messages.hh"
#include "utils/ref_frames/include/tree_links.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::MassPointLinks](#)
Encapsulates the links between mass points.

Namespaces

- [jeod](#)
Namespace jeod.

9.15.1 Detailed Description

Define the class MassPointLinks, the class that encapsulates the links between mass points.

9.16 mass_point_mass_inertia.cc File Reference

Compute the inertia tensor of a point mass.

```
#include "../include/mass.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.16.1 Detailed Description

Compute the inertia tensor of a point mass.

9.17 mass_point_state.cc File Reference

Define basic methods for the MassPointState class.

```
#include "utils/math/include/matrix3x3.hh"  
#include "utils/math/include/numerical.hh"  
#include "utils/math/include/vector3.hh"  
#include "../include/mass_point_state.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.17.1 Detailed Description

Define basic methods for the MassPointState class.

9.18 mass_point_state.hh File Reference

Define the class MassPointState, which defines the state – position and orientation – of a MassPoint.

```
#include "class_declarations.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/quaternion/include/quat.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/math/include/macro_def.hh"
#include "utils/math/include/macro_undef.hh"
```

Data Structures

- class [jeod::MassPointState](#)
Defines the state – position and orientation – of a [MassPoint](#).

Namespaces

- [jeod](#)
Namespace jeod.

9.18.1 Detailed Description

Define the class MassPointState, which defines the state – position and orientation – of a MassPoint.

9.19 mass_print_body.cc File Reference

Define MassBody::print_body.

```
#include <cstdio>
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include "../include/mass.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.19.1 Detailed Description

Define MassBody::print_body.

9.20 mass_print_tree.cc File Reference

Print out the MassTree members' data to a given file.

```
#include <cstdint>
#include <cstdio>
#include "utils/message/include/message_handler.hh"
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.20.1 Detailed Description

Print out the MassTree members' data to a given file.

9.21 mass_properties.hh File Reference

Define the class MassProperties.

```
#include "class_declarations.hh"
#include "mass_point.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::MassProperties](#)

Defines mass properties – mass and inertia tensor.

Namespaces

- [jeod](#)

Namespace jeod.

9.21.1 Detailed Description

Define the class MassProperties.

9.22 mass_properties_init.cc File Reference

Define methods for the MassPropertiesInit class.

```
#include "../include/mass_properties_init.hh"
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/message/include/message_handler.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.22.1 Detailed Description

Define methods for the MassPropertiesInit class.

9.23 mass_properties_init.hh File Reference

Define class MassPropertiesInit, which initializes a mass properties object.

```
#include "class_declarations.hh"
#include "mass_point_init.hh"
#include "utils/orientation/include/orientation.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::MassPropertiesInit](#)

Contains data used to initialize a mass model object.

Namespaces

- [jeod](#)

Namespace jeod.

9.23.1 Detailed Description

Define class MassPropertiesInit, which initializes a mass properties object.

9.24 mass_reattach.cc File Reference

Define MassBody::reattach.

```
#include "utils/message/include/message_handler.hh"
#include "../include/mass.hh"
#include "../include/mass_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.24.1 Detailed Description

Define MassBody::reattach.

9.25 mass_update.cc File Reference

Define MassBody::update_mass_properties.

```
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/ref_frames/include/tree_links_iterator.hh"
#include "../include/mass.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.25.1 Detailed Description

Define MassBody::update_mass_properties.

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